NTU / SC6122 / GROUP 5

# Credit Card Default Prediction

Risk, interpretation and decision costs

Emerging Topics in FinTech


THE TEAM

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Contribution: 25% each

# How we split and use the data

30,000 clients • 23 predictors • 22.12% default

Categories learned within each training fold

Validation selects policies; historical test reports outcomes

## Partition and purpose

| | |
|---|---|
| **Fitting**<br>Five-fold CV and final fitting | 18,000 |
| **Validation**<br>Select decision thresholds | 6,000 |
| **Test**<br>Evaluate frozen decisions | 6,000 |

Source: data/splits; protocol_frozen.json; results/final/audit.json

# Logistic regression: the linear reference

L2 model + fold-trained scaling + one-hot categories

12 declared settings; CV selected C=0.1, unweighted

Original LR notebook was empty; original process unverified

**AP 0.4967**

Baseline AP 0.4970

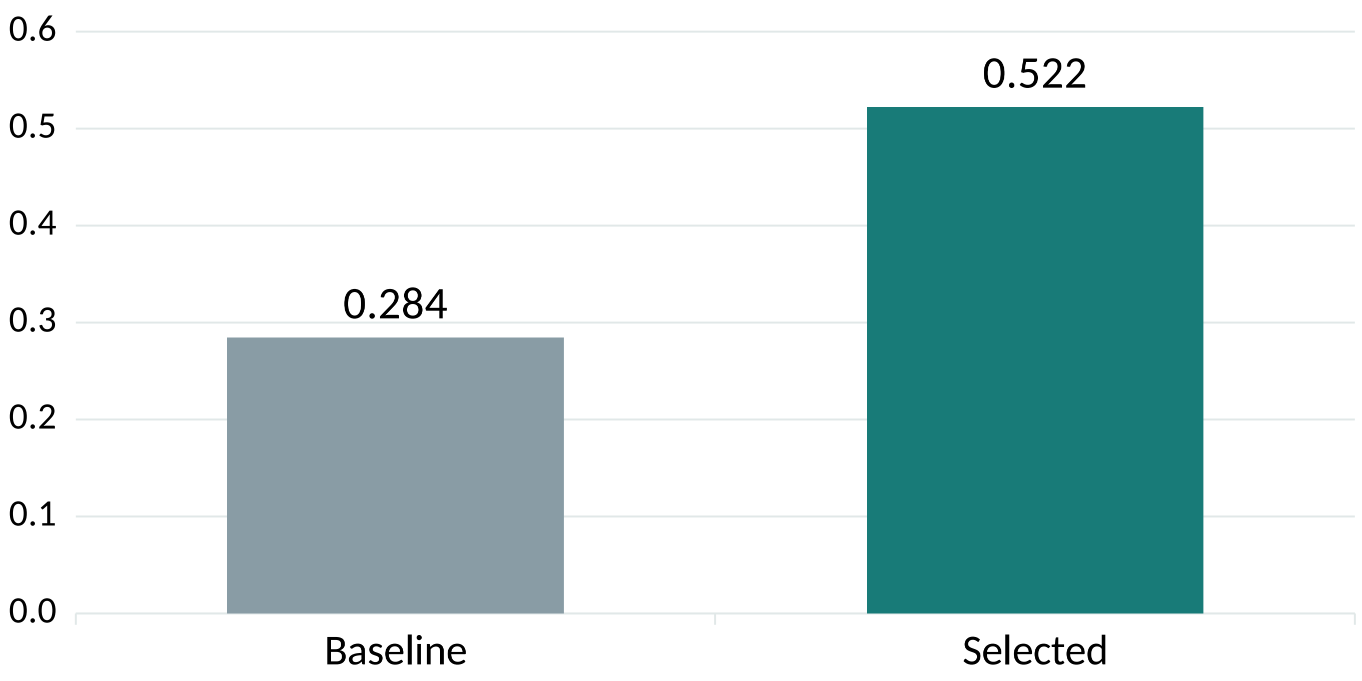
No ranking improvement from selection

LR* is a separate post-hoc supplement

Source: results/final/model_comparison.csv

# Decision tree: controlling overfitting

## Test average precision




**21 configurations: depth × minimum leaf size**

Unrestricted baseline fits training folds almost perfectly

Selected minimum leaf size: 100 observations

A large baseline gap signals overfitting.

CV selection improves this tree's ranking.

Source: results/final/model_comparison.csv

# Reading the tree’s first decision

Root rule: recent repayment status PAY_0 ≤ 1.5

Selected tree: depth 18, with 134 leaves

Node proportions describe this fitting sample

## Recent repayment status

**PAY_0 ≤ 1.5**

Yes · 16.51%

No · 70.13%

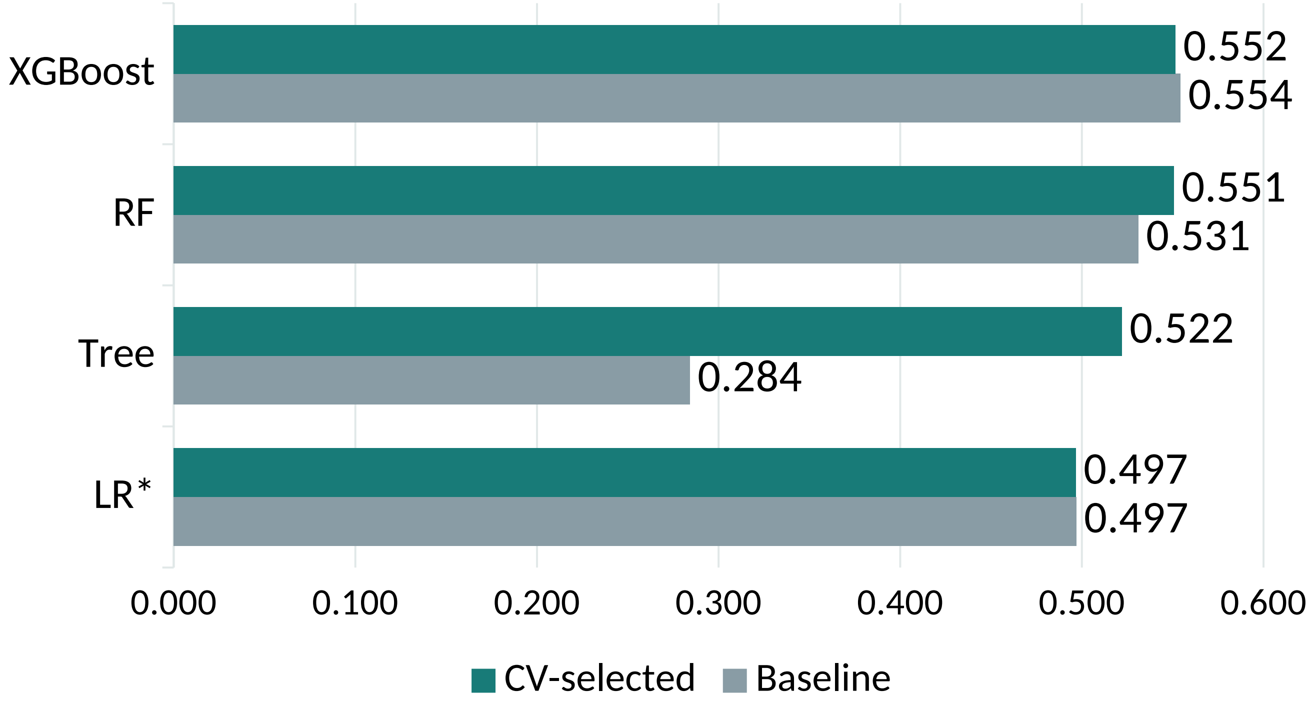
Default proportion in each fitting-sample branch.

These are associations, not causal effects.

Source: saved decision-tree model; integration audit/report evidence.

# Ranking performance across four models

## Test average precision




Source: results/final/model_comparison.csv

**Decision tree gains most on this split**

RF and XGBoost have the highest selected AP

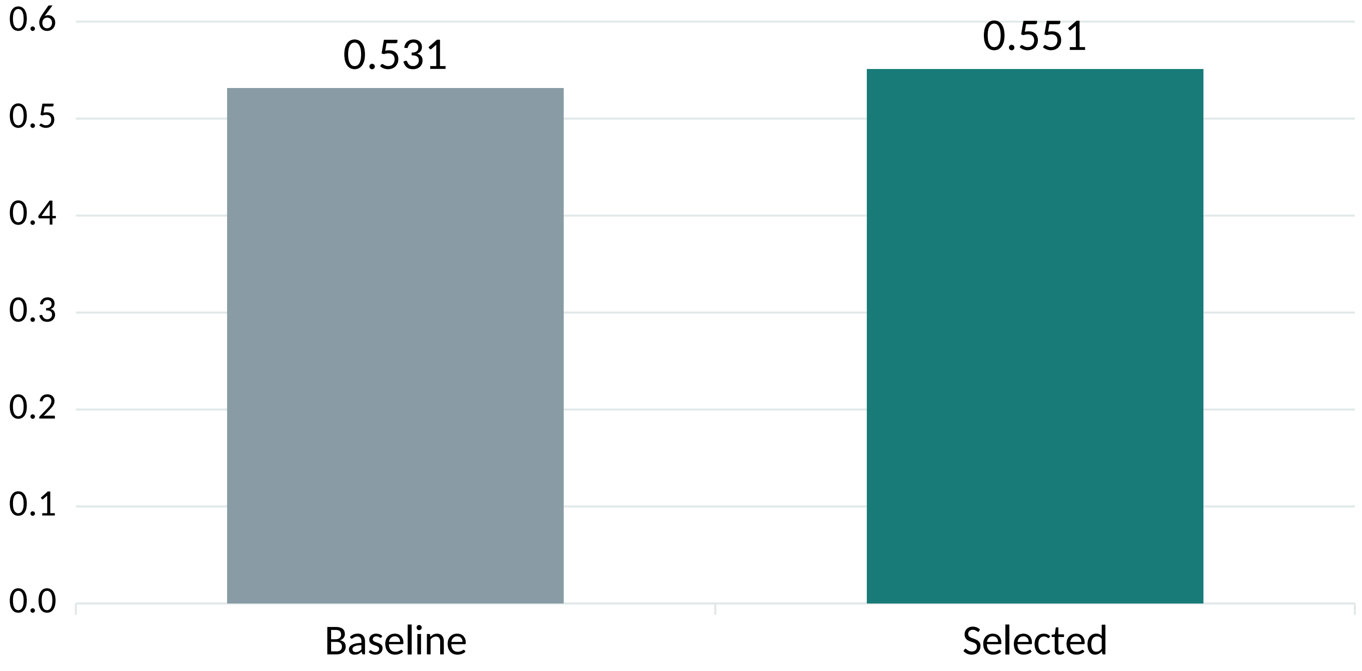
CV selection precedes test comparison

Baseline and selected models are both retained.

Small ensemble differences do not establish superiority.

# Random forest: averaging and regularization

## Test average precision




Source: results/final/model_comparison.csv

**Baseline + 23 seeded candidate settings**

Selected: 500 trees, depth 8, minimum leaf 2

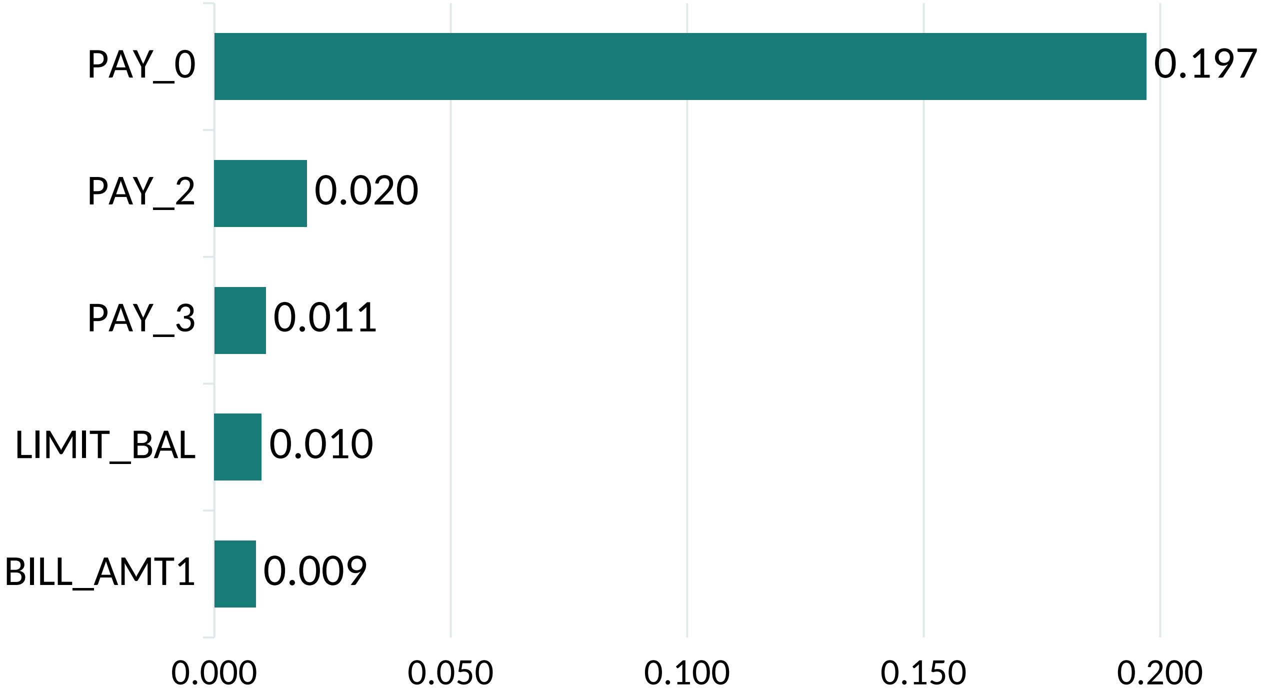
Positive-class weight 3; feature fraction 0.5

Averaging reduces dependence on one tree.

Weights also change the score distribution.

# Random forest: which features matter?

## Mean decrease in validation AP




**PAY_0 has the largest validation AP decrease**

Permutation measures predictive reliance

Correlated variables can substitute for each other

Five saved shuffles per feature.

Predictive association, not causality.

Source: results/rf/validation_importance.csv

# Random forest: two illustrative errors

At 0.5: 637 missed defaults; 573 false alarms

Missed default · row 982

Score 0.1012; actual default = 1

False alarm · row 28747

Score 0.9415; actual default = 0

## Why the predictions can fail

Missed default: PAY_0 = -2

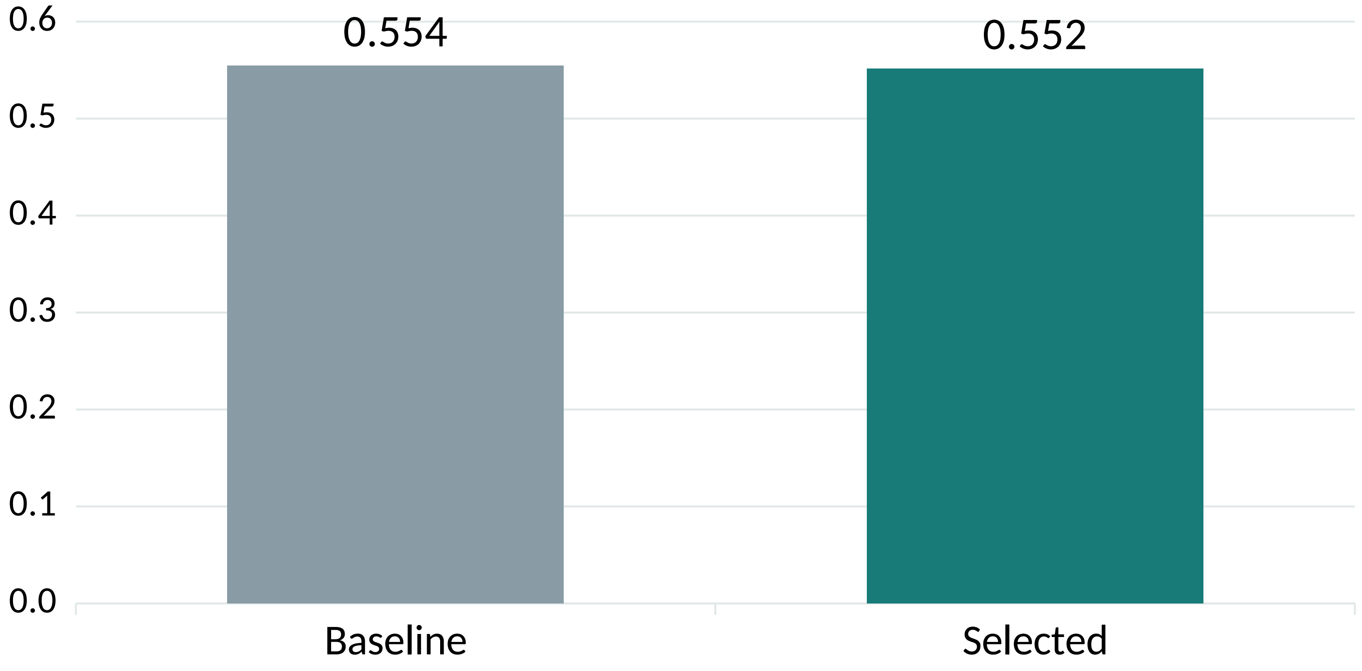
False alarm: PAY_0 = 3

Illustrative extreme errors, not typical clients.

Source: results/rf/misclassification_cases_05.csv

# XGBoost: no gain in test ranking

## Test average precision




**24 configurations; select by CV AP**

Selected: 150 trees, depth 4, rate 0.03

Selected ROC-AUC: 0.7762

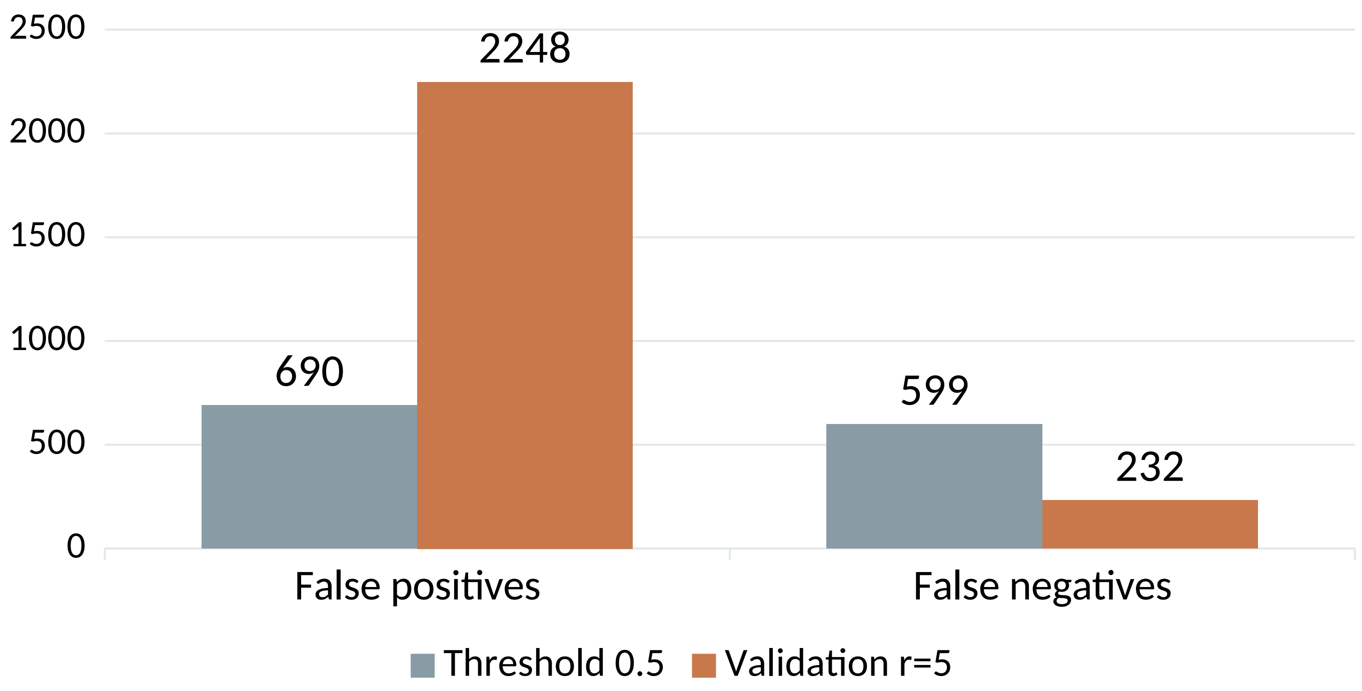
Tuned − baseline AP: -0.0028

Conditional 95% interval: [-0.0106, 0.0054]

Source: results/final/model_comparison.csv

# XGBoost: fewer misses, more reviews

## Test confusion trade-off




Source: results/final/model_comparison.csv

## Validation r=5 cutoff: 0.3150

Recall 54.9% → 82.5%; alerts 55.7%

Cost 3,685 → 3,408: 7.5% reduction

Same ranking scores, different action policy.

Review costs are hypothetical; capacity is unmeasured.

# Conclusions & Q&A

Ranking, explanations, and workload need separate evidence

No clear ensemble winner from these small differences

Next: fresh temporal data, calibration, subgroup errors

## Thank you

Questions & discussion

Lei Peng · Zhang Hanyu
Zhou Xinzhe · Xu Yiqun

Historical educational study; no causal or deployment-readiness claim.

# Backup A • AP, ROC-AUC and threshold metrics

AP = sum of recall increments × precision

ROC-AUC measures positive/negative score ordering

Precision, recall, F1 and accuracy require a threshold

## Two different definitions

Saved continuous scores supply ranking metrics.

Changing the cutoff changes decisions only.

Source: scikit-learn average_precision_score documentation; part4/evaluation.py

# Backup B • Cost ratios and fixed-prediction uncertainty

Predeclared ratios: r = 1, 3, 5, 10

Validation minimum cost; ties prefer largest cutoff

1,000 stratified paired bootstrap resamples

## What the intervals cover

Fixed models / thresholds / class counts

Excludes fitting / selection uncertainty

Excludes population / prevalence shifts

Source: frozen protocols; integration/uncertainty.py; bootstrap_intervals.csv

# Backup C • Data provenance and evidence boundaries

UCI cleaning reproduced exactly from downloaded source

EDUCATION 0/5/6 → 4; MARRIAGE 0 → 3

Retained duplicate vectors can span partitions

## The limits of historical evidence

Original LR process remains unverified.

LR* source was committed before its one run.

No new unseen-test claim is made.

Sources: results/final/source_audit.json; integration/LOGISTIC_PROVENANCE.md

# Backup D • Sources, responsibilities and reproducibility

UCI: doi.org/10.24432/C55S3H

scikit-learn: average_precision_score documentation

XGBoost paper: doi.org/10.1145/2939672.2939785

## Member contributions · 25% each

**Lei Peng · G2509090C**

Part 1 — Data and logistic regression

**Zhang Hanyu · G2509091L**

Part 2 — Decision tree and overfitting

**Zhou Xinzhe · G2509033F**

Part 3 — Random forest and error analysis

**Xu Yiqun · G2509092H**

Part 4 — XGBoost and threshold trade-offs

Sources: Group 5 repository, frozen protocols, row-level predictions and integration audit.